

Web Engineering 2 - Project Description

Key Features

In this final project, your team will develop a dynamic full-stack web application. You are free to choose the theme of your website, or you may continue improving your existing web project.

The aim is not only to create a nice-looking website, but to demonstrate your ability to design and implement a complete web application with:

- backend functionality,
- database storage,
- user authentication,
- CRUD operations,
- interaction between users and content,
- search or filtering,
- clear MVC structure,
- and a usable, responsive interface.

Required Key Features

1. User Registration and Authentication:
 - 1.1. Users should be able to:
 - 1.1.1. Register,
 - 1.1.2. log in,
 - 1.1.3. log out,
 - 1.1.4. access protected pages only after login.
 - 1.1.5. Passwords should not be stored as plain text.
2. User Profiles:
 - 2.1. Each user has a profile displaying their username, bio, profile picture, and a list of items created by the user.
 - 2.2. Users can edit their profiles.

“The exact profile content depends on your project theme”
3. Main Content Creation: CRUD
 - 3.1. Your website must have a main content type. Users should be able to:

- 3.1.1. create,
 - 3.1.2. view,
 - 3.1.3. edit,
 - 3.1.4. delete.
- 4. Content Details:
 - 4.1. Each main item should include meaningful fields, such as:
 - 4.1.1. title or name,
 - 4.1.2. description or content,
 - 4.1.3. category,
 - 4.1.4. tags,
 - 4.1.5. image or featured image,
 - 4.1.6. author/creator,
 - 4.1.7. date created or updated.
 - “The exact fields should fit your chosen theme”
- 5. Comments and Interactions:
 - 5.1. Users should be able to interact with content through comments, reviews, or feedback.
- 6. Search and Filtering:
 - 6.1. Your website must allow users to search or filter content.
- 7. Likes, Ratings, Favorites, or Bookmarks
 - 7.1. Your website should include at least one additional interaction feature, such as:
 - 7.1.1. like,
 - 7.1.2. rating,
 - 7.1.3. favorite,
 - 7.1.4. bookmark,
 - 7.1.5. save for later.
 - “Choose the interaction that fits your theme”
 - 7.2. Include social media sharing buttons for readers to easily share posts on platforms like Twitter and Facebook.
- 8. MVC Structure
 - 8.1. Your project should follow the Model-View-Controller pattern.
 - 8.2. You should clearly separate:
 - 8.2.1. models: database structure and data logic,
 - 8.2.2. views: pages/templates/user interface,
 - 8.2.3. controllers: request handling and application logic,
 - 8.2.4. Routes: URL structure.

- 9. Responsive Design:
 - 9.1. Ensure that the platform is fully responsive for optimal user experience on various devices.
- 10. Unique URL for each item
 - 10.1. Each item should have its own page with a unique URL.
 - 10.2. It is preferable to have user-friendly URLs.
- 11. Accessibility:
 - 11.1. Adhere to web accessibility guidelines to ensure your website is accessible to users with disabilities.
 - 11.2. Properly label form elements and provide alternative text for images, among other accessibility considerations.
- 12. UI Guidelines:
 - 12.1. Follow UI guidelines for typography and layouts.
 - 12.2. Choose fonts, font sizes, and spacing that align with your selected theme while maintaining readability.
 - 12.3. Easy Navigation:
 - 12.3.1. Implement clear and intuitive navigation methods on your website.
 - 12.3.2. Use navigation menus, breadcrumbs, or other appropriate techniques to facilitate easy navigation.

Recommended Technology Stack

1. **Backend:** Node.js with Express.js
2. **Database:** You can choose between a Relational database (e.g. MySQL or SQLite) or a noSQL database e.g. MongoDB for data storage.
3. **Frontend:** You have the option to use a frontend framework (e.g., React, Angular, or Vue.js) or develop the user interface using EJS templates.
4. **Authentication:** Passport.js or similar library
5. **Rich Text Editor:** Use a suitable library like Draft.js or Quill.js.
6. **MVC Pattern:** Structure the application following the Model-View-Controller (MVC) pattern for maintainability and scalability.
7. **Version Control:** Use GitHub for version control and collaborative development.
8. **Deployment:** Deploy the application on a cloud platform (e.g., Heroku, AWS, Vercel or Azure).



Grading Criteria

The project will be evaluated based on the following criteria:

1. The quality and functionality of the blogging platform, database design, adherence to the MVC pattern, deployment success, code quality, and the ability to customize the frontend.
2. Usability and Overall Design: Visual appeal and creativity of the website design.
3. Project Presentation.

Project Deliverables

1. A code repository:
 - a. source code,
 - b. clear folder structure,
 - c. README file,
 - d. installation/running instructions,
 - e. database setup instructions.
2. A live website: Host your website on a web server and provide the URL for the website.

Presentation

At the end of the project (last lecture), the whole team will present the website and explain the design choices, implementation techniques, and the overall user experience.